

THE SAFETY NET

Handling Mistakes in Python

Python Programming | Grade 4

Imagine This...

Imagine you're going to school in a colorful rickshaw, excited for the day, and suddenly... **PUNCTURE!** [cite: 380, 381]

Visual: A bright rickshaw with a flat tire on a busy Indian street. [cite: Page 2 Illustration]

The tire goes flat and your ride stops! Uh-oh! What happens now? [cite: 381]

Unexpected problems like this can happen anytime—even to our programs! In coding, we call these surprise problems **Exceptions**. [cite: 383, 384]

What is an Exception?

An Exception is an unexpected problem that stops our program from working properly. [cite: 385]

Visual: A smiley warning sign representing a manageable error. [cite: Page 3 Illustration]

Remember This! [cite: 387]

- **Exception** = Surprise problem. [cite: 388]
- It can make the program stop suddenly. [cite: 389]
- But don't worry, we can **handle** it! [cite: 390]

Just like a rickshaw puncture stops your ride to school, an exception stops your code! [cite: 386]

What is 'try'?

The try block is like testing if you have enough milk at home before you start making **Chai**. [cite: 391, 393]

Visual: A child in a kitchen pouring milk to prepare tea. [cite: Page 4 Illustration]

We put code that might cause a problem inside 'try' to test it safely! [cite: 394]

```
try:  
    make_Tea() # We try this first!
```

What is 'except'?

If our first plan doesn't work, we need a **Backup Plan**. That is what `except` is for! [cite: 400, 401]

Visual: A child running to a local Kirana store. [cite: Page 5 Illustration]

The Kirana Analogy: If the milk is finished at home (the error), our backup plan is to go to the **Kirana store** to buy more! [cite: 402]

```
except:  
    go_to_kirana_store()
```

ZeroDivisionError

Trying to divide by zero is like trying to share sweets when there are **zero** friends—it just doesn't work! [cite: 410, 411]

Visual: A child looking confused at a calculator with a warning. [cite: Page 6 Illustration]

```
try:
    result = number / 0
except ZeroDivisionError:
    print("Oopsie! You can't divide by zero.")
```

With try and except, our calculator stays friendly instead of crashing! [cite: 417, 418]

Graceful Failure

Graceful failure means saying "Oopsie!" nicely instead of stopping suddenly. [cite: 442, 445]

- **ATM Example:** If an ATM runs out of cash, it shows a "No cash" message instead of switching off! [cite: 443]
- **Power Cut Example:** When the power goes out, the **Inverter** switches on to save the day! [cite: 444]

Visual: An ATM showing a friendly "Sorry, no cash available" message. [cite: Page 9 Illustration]

Remember: Errors are just chances to learn! Keep practicing your safety nets!
[cite: 456, 457]